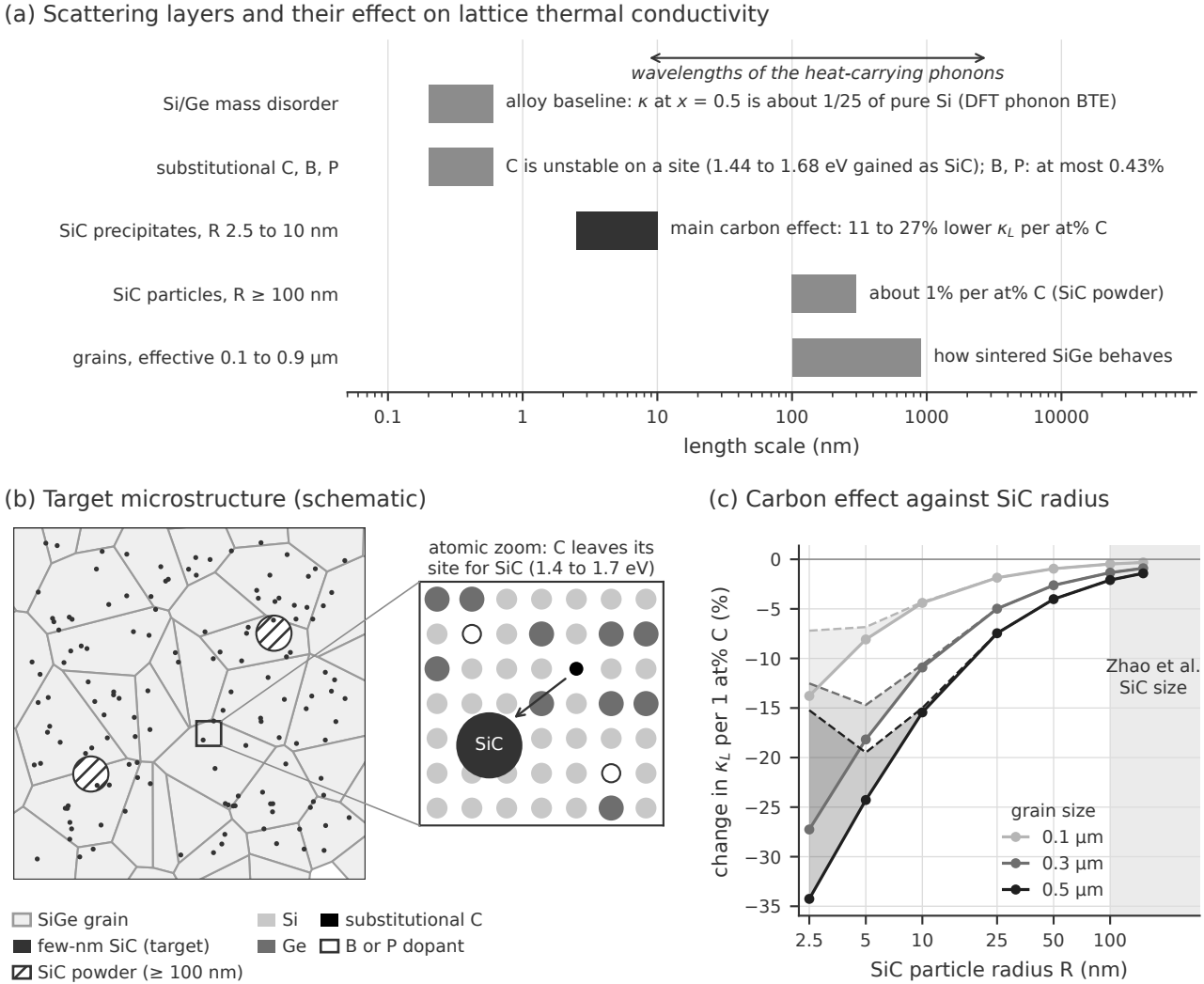

(a) Scattering layers and their effect on lattice thermal conductivity
wavelengths of the heat-carrying phonons
Si/Ge mass disorder
alloy baseline: κ at x = 0.5 is about 1/25 of pure Si (DFT phonon BTE)
substitutional C, B, P
C is unstable on a site (1.44 to 1.68 eV gained as SiC); B, P: at most 0.43%
SiC precipitates, R 2.5 to 10 nm
main carbon effect: 11 to 27% lower κL per at% C
SiC particles, R ≥ 100 nm
about 1% per at% C (SiC powder)
grains, effective 0.1 to 0.9 µm
how sintered SiGe behaves
0.1
1
10
100
1000
10000
length scale (nm)
(b) Target microstructure (schematic)
atomic zoom: C leaves its site for SiC (1.4 to 1.7 eV)
SiC
SiGe grain
Si
substitutional C
few-nm SiC (target)
Ge
B or P dopant
SiC powder (≥ 100 nm)
(c) Carbon effect against SiC radius
change in κL per 1 at% C (%)
0
−5
−10
−15
−20
−25
−30
−35
Zhao et al.
SiC size
grain size
0.1 µm
0.3 µm
0.5 µm
2.5
5
10
25
50
100
SiC particle radius R (nm)